



Office of the President

Rebecca Cantor, Ph.D.
President

September 10, 2026

Dr. F. David Winter, President
The Discovery Foundation
4514 Cole Ave, Ste 600
Dallas, TX 75205

Dear Dr. Winter,

On behalf of Austin College, I am pleased to submit the enclosed proposal requesting \$30,946 for Dr. Md Mohsan Khudri's project, "MOSAIC-TX: Mapping and Optimizing Obstetric Care Access in Texas through a Data-Driven Approach."

Maternal mortality and severe maternal morbidity represent significant and largely preventable public health challenges in Texas. With nearly half of Texas counties classified as maternity care deserts and substantial disparities in access to prenatal, delivery, and postpartum services, there is an urgent need for evidence-based strategies that help policymakers allocate limited resources where they can have the greatest impact. By combining advanced geographic analysis, optimization modeling, and machine learning techniques to identify the most effective approaches for expanding obstetric care capacity across the state, this project will directly address that need.

Led by Dr. Khudri, Assistant Professor of Economics at Austin College, the interdisciplinary team brings expertise in health economics, healthcare access, machine learning, spatial analytics, and software development. Together with Dr. Navid Mohammad Imran of William Paterson University and Dr. Alamgir Hosain of EnviTrace, the team will develop and validate a statewide census-block-level measure of obstetric care access, create cost-aware facility siting models, and produce a free, open-source decision-support platform for state and local health planners. Through an accessible planning platform that translates complex analyses into actionable guidance, the project has the potential to improve healthcare access for women in Texas and serve as a model nationwide.

I appreciate The Discovery Foundation's past and present support for our remarkable faculty and undergraduates. Thank you for considering this proposal. Please contact me if I can provide more information.

Sincerely,

A handwritten signature in blue ink, appearing to read 'RC', followed by a long horizontal flourish.

Rebecca Cantor, Ph.D.
President

MOSAIC-TX: Mapping and Optimizing Obstetric Care Access in Texas through a Data-Driven Approach

Significance

Maternal mortality remains a major public health concern in the United States. In 2021, the national maternal mortality ratio (MMR) reached 32.9 deaths per 100,000 live births (Hoyert, 2023). Maternal mortality is also a major concern in Texas, where substantial disparities persist across communities. In Texas, the MMR increased from 17.0 deaths per 100,000 live births in 2018 to 27.7 in 2020, an increase of approximately 63 percent. The Texas Maternal Mortality and Morbidity Review Committee also determined that 80 percent of pregnancy-related deaths reviewed for 2020 were preventable (Texas Maternal Mortality and Morbidity Review Committee & Texas Department of State Health Services, 2024). These patterns underscore the importance of timely access to appropriate obstetric care.

Geographic access remains a major challenge in Texas. More than 46 percent of Texas counties are classified as maternity care deserts (MCDs)¹ - meaning they lack a birthing hospital or birth center and obstetric providers (Stoneburner et al., 2026). Women living in these areas may need to travel substantial distances for maternity care. Texas also has higher rates of inadequate prenatal care than the national average (Stoneburner et al., 2026). The problem is particularly important in rural communities, where financial pressures and workforce shortages continue to contribute to obstetric unit closures (Topchik et al., 2024; U.S. Government Accountability Office, 2022). These closures can further increase travel burdens and reduce timely access to care.

This project addresses these challenges by moving beyond broad county-level measures of maternity care access. We will first establish a validated, census-block-level measure of access based on actual road travel. We will then use Integer Linear Programming (ILP) and Reinforcement Learning (RL) models that account for facility costs and budget constraints to identify expansion strategies that maximize population coverage. The models will account for both population demand and facility capacity. They will also distinguish between constructing new facilities and restoring obstetric services in existing hospitals. This distinction is important because restoring services in existing hospitals may provide a more feasible option in some underserved communities. Finally, we will translate these models into a free, open-source software package. State and county planners will be able to evaluate alternative budgets, access targets, and facility-expansion scenarios.

Improving geographic access may reduce delays in obtaining maternity care and support timelier prenatal, delivery, and postpartum services. The economic implications are also substantial. The societal cost of nine selected maternal morbidity conditions associated with pregnancies and births in the U.S. in 2019 was estimated at \$32.3 billion from conception through five years postpartum (O’Neil et al., 2022). By identifying where additional obstetric capacity could produce the greatest improvement in access, this project will provide Texas decision-makers with an evidence-based framework for allocating limited maternal health resources. The software package will serve as an interactive planning tool, allowing recommendations to be updated as budgets, populations, and facility availability change. In this way, the project directly supports The Discovery Foundation’s mission to advance women’s health and wellness through research and practical initiatives.

Investigator(s)

Md Mohsan Khudri, PhD, Assistant Professor of Economics at Austin College, will serve as Principal Investigator. Dr. Khudri is a health economist with training in economics and applied statistics and expertise in health disparities, women’s health, healthcare access, causal inference, and

¹A county is classified as a MCD if it does not have a birthing hospital or birth center, as well as obstetric providers such as obstetrician-gynecologists, certified nurse midwives, certified midwives, and family physicians who deliver babies (Stoneburner et al., 2026).

machine learning. His research has appeared in peer-reviewed journals including *Economic Modelling*, *Applied Economics*, *Public Health*, and *PLOS ONE*. He will provide overall scientific leadership, direct the study design and analytical framework, supervise the project team, and oversee data management, project milestones, compliance, and policy translation.

Navid Mohammad Imran, PhD, Assistant Professor of Computer Science at William Paterson University of New Jersey, will serve as Co-Principal Investigator. Dr. Imran specializes in machine learning, combinatorial optimization, and dynamic algorithmic modeling, with publications in venues including *IEEE Transactions on Intelligent Transportation Systems* and the *IEEE/RSJ International Conference on Intelligent Robots and Systems*. He will direct the technical development of the ILP and RL models and contribute to computational implementation and model validation.

Alamgir Hosain, PhD, Research Scientist at EnviTrace in Santa Fe, New Mexico, will serve as Co-Principal Investigator. Dr. Hosain brings expertise in computational geosciences, spatial data integration, routing systems, and scientific software development. His preliminary contributions include improving the facility registry and developing the 11-million-node road network and routing framework. He will support data integration, demand-surface development, geographic coverage analysis (Hosain et al., 2026), and development of the open-source software package.

Innovation

As part of our preliminary investigation, we have developed several innovations to improve the measurement of access to obstetric care in Texas. Rather than relying on county-level averages or straight-line distances, we calculated road-based travel distances from all 668,757 Texas census blocks to active obstetric facilities (Figure 1). Figure 1 illustrates the geographic variation in access to obstetric and NICU-capable care across Texas. It also shows the geographic distribution of counties with existing hospitals that currently do not provide obstetric services.

These estimates were weighted by the population of women aged 15-44 and evaluated across multiple travel-distance thresholds of 25, 35, and 50 miles. We also developed an efficient statewide routing method that eliminates reliance on costly commercial routing services and enables calculation of travel times across the entire Texas road network. Finally, we created an entity-resolution procedure that matched facility records using license identifiers, addresses, and standardized name matching.

Existing facility-location studies have primarily used traditional optimization or statistical methods (Haley et al., 2024; Meredith et al., 2024; Snug et al., 2025). While these approaches are effective for identifying optimal locations based on fixed assumptions, they are less suitable for planning when population needs, funding, and hospital services fluctuate over time. To address this, we will model multi-year obstetric facility expansion as a sequential decision problem under uncertainty, employing RL. Unlike static optimization models, the RL framework can update facility-placement decisions as conditions evolve (Kaelbling et al., 1996). This enables us to evaluate whether a more adaptive strategy improves access when populations shift, budgets change, or obstetric units unexpectedly close. We will benchmark RL recommendations against deterministic optimization to assess whether this flexibility leads to meaningful improvements in access and resource use.

We will also integrate ILP with Two-Step Floating Catchment Area (2SFCA) capacity modeling to make facility-expansion recommendations more realistic (Luo & Wang, 2003). The ILP framework will differentiate between building new obstetric facilities and restoring services in existing hospitals. Our preliminary analysis identified 78 counties with hospitals that do not currently provide obstetric care, presenting opportunities to restore lower-cost obstetric services. Rather than relying solely on proximity, the 2SFCA component will consider both facility capacity and population demand as indicators of access. By incorporating the average installation cost and a facility's clinical capacity, the model can identify expansion strategies that improve geographic access while remaining feasible within resource constraints.

Finally, we will translate the optimization models into a free, open-source, web-based planning tool designed for state and county decision-makers. The platform will enable users to adjust budget constraints, access targets, and facility scenarios to generate tailored siting recommendations. It will produce maps and exportable tables to support planning and resource allocation decisions. This tool is intended to bridge the gap between academic research and policy implementation by allowing non-technical users to apply the models as local needs and available resources evolve. Making the platform open source will further enhance transparency and allow the framework to be adapted for use beyond Texas.

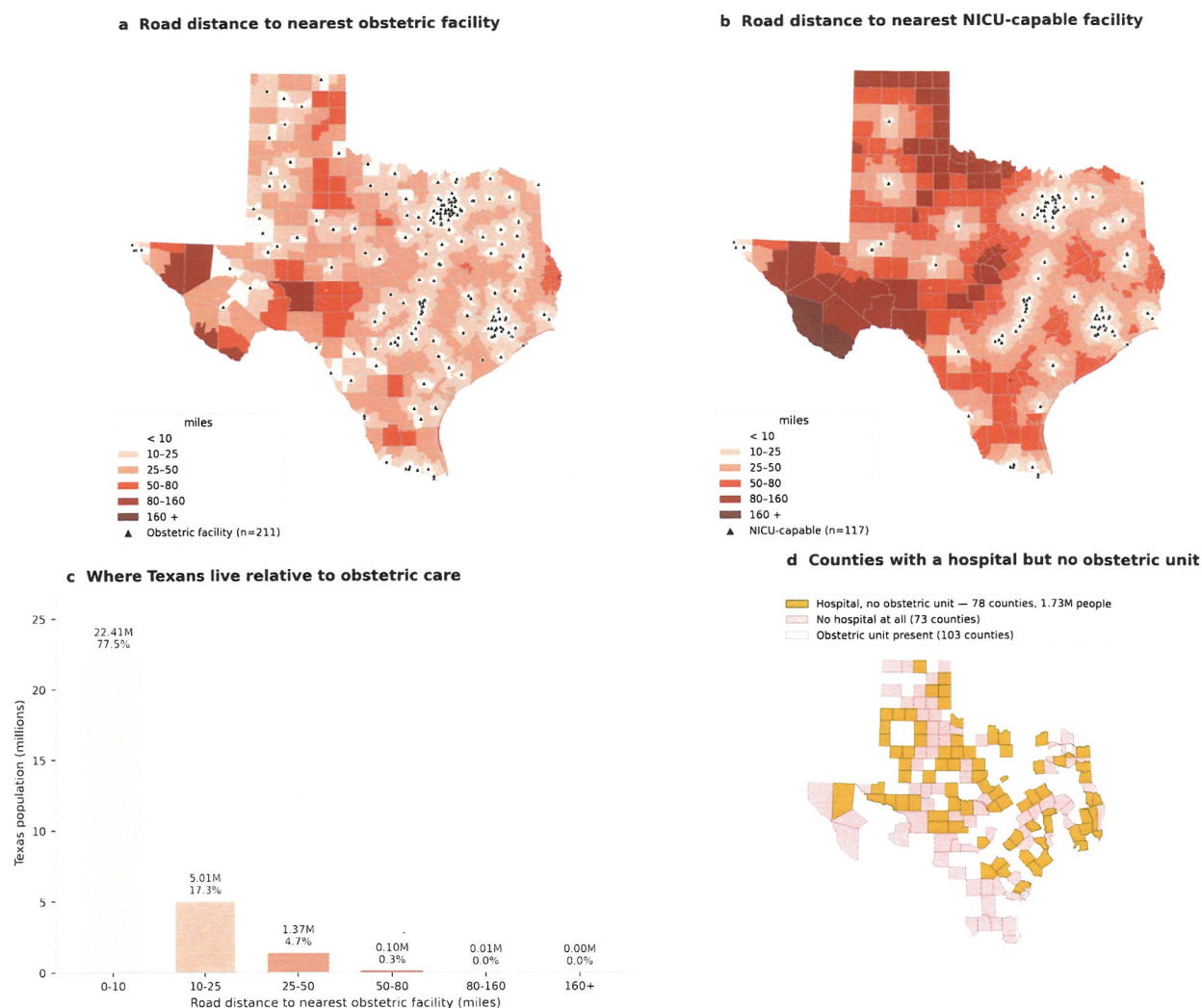


Figure 1. Obstetric care access across Texas: (a) road distance to the nearest obstetric facility; (b) road distance to the nearest NICU-capable facility; (c) population by distance to the nearest obstetric facility; and (d) counties with a hospital but no obstetric unit (gold), counties with no hospital (pink), and counties with an obstetric unit (white).

Approach

This project has three aims: to establish and validate a census-block-level baseline of obstetric care access across Texas; to develop cost-informed ILP and dynamic RL models for optimal facility siting; and to translate these models into an interactive, open-source decision-support platform for policymakers and health planners.

Aim 1: Validate and Refine the Statewide Access Baseline (Months 1–3)

Building on our completed census-block routing pipeline, we will finalize the statewide baseline for obstetric care access using American Community Survey (ACS) data for women aged 15–44, HRSA facility locations, and Centers for Medicare & Medicaid Services (CMS) provider of Services archives. We will validate free-flow road travel times by comparing them to results from a commercial routing engine, using a stratified sample of approximately 500 routes. Block-group demographic estimates will be allocated to the census-block level, and historical CMS records will be used to identify obstetric unit closures. Access will be assessed based on travel distance thresholds of 25, 35, and 50 miles. These measures will be calculated separately for general obstetric care and for Level III/IV²³ NICU-capable facilities.

Aim 2: Develop Cost-Aware ILP and Dynamic RL Siting Models (Months 4–7)

We will develop an ILP framework to identify facility locations that maximize population coverage within a specified budget. The ILP framework will account for cost considerations and facility capacity using the 2SFCA method. It will distinguish between constructing new facilities and restoring obstetric services in existing hospitals across 78 counties. This distinction will allow the model to identify potentially lower-cost strategies for expanding access.

In parallel, we will develop an RL model to identify facility placement strategies that continuously adapt to changing conditions. We will formulate the problem as a sequential decision-making process that accounts for population shifts, facility closures, and persistent coverage gaps. The model's state will capture up-to-date access conditions, including areas that remain underserved and facilities already added. At each stage, the model will determine the most effective location for a new facility. The reward function will incentivize maximizing coverage while penalizing unnecessary facility placements and excessive resource use. We will then benchmark the resulting RL policy against the deterministic ILP solution to assess whether this adaptive approach yields better coverage or efficiency.

Aim 3: Deploy the Interactive Decision-Support Platform (Months 8–12)

To fulfill our deliverable innovation, we will embed the ILP and RL optimization engines into a free, open-source web application designed for state and county planners. This software package will allow stakeholders to adjust coverage parameters and budget constraints to identify optimal facility expansions. Users can weigh the data by either the reproductive-age female population or the total population, and target either general obstetric or NICU-capable facilities. The software will then generate exportable tables and maps. All project code will be released under an open license to ensure that the framework can be adapted for use across all 50 states.

² Level III maternal care provides subspecialty and critical care for complex maternal and obstetric conditions, with 24-hour access to specialized clinical support (“Levels of Maternal Care,” 2019; 25 Tex. Admin. Code § 133.208).

³ Level IV maternal care includes all Level III capabilities and provides advanced medical and surgical care for the most complex maternal conditions, including critically ill patients (“Levels of Maternal Care,” 2019; 25 Tex. Admin. Code § 133.209).

There are several potential challenges in this project. These include lags in CMS registry data, unmodeled traffic congestion, border-county edge effects, and the computational demands of RL training. To address data quality concerns, we will cross-check CMS flags against Texas Department of State Health Services (DSHS) maternal designations and American Hospital Association (AHA) surveys. We will also calibrate free-flow travel times using samples from a commercial routing API such as Google Maps. To address border-county effects, we will extend the road network into neighboring states. If needed, we will use cloud computing infrastructure for RL training. Our cost-aware ILP framework will remain a fully functional alternative if the experimental RL policy does not outperform static optimization.

Resources

Dr. Khudri, as Principal Investigator, will receive institutional support from Austin College throughout the grant period. The College will provide access to analytics laboratory facilities, seminar space, secure data storage, computing resources, and routine IT support. Administrative staff will assist with scheduling, workshop coordination, purchasing, budgeting, reporting, procurement, and other grant-management activities. Austin College will also provide guidance on IRB review, data governance, and data use agreements to support compliance with institutional and regulatory requirements.

The College will support the planned effort of the PI and co-investigators and provide in-kind contributions through facility use, computing resources, and staff time. In addition, Austin College will provide pre-award and post-award administrative support and assist with planning for the long-term sustainability of the project beyond the grant period.

References

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Project Budget	Year 1 (01/01/27-12/31/27)
Summer Stipend for Faculty (PI)	\$ 7,500
Summer Stipends for 2 Co-PIs (Faculty) @ \$6000 each	\$ 12,000
Semester Stipends for 1 Student (1,200 each semester)	\$ 1,200
Stipend Taxes and Benefits	\$ 1,433
Publication Cost	\$ 3,000
Conference	\$ 1,500
Data & Software	\$ 1,500
Total Direct Costs	\$ 28,133
Indirect Costs (10%)	\$ 2,813
Total	\$ 30,946

Note: This project will not generate revenue.